

ChIP-on-Chip and Differential Location Analysis

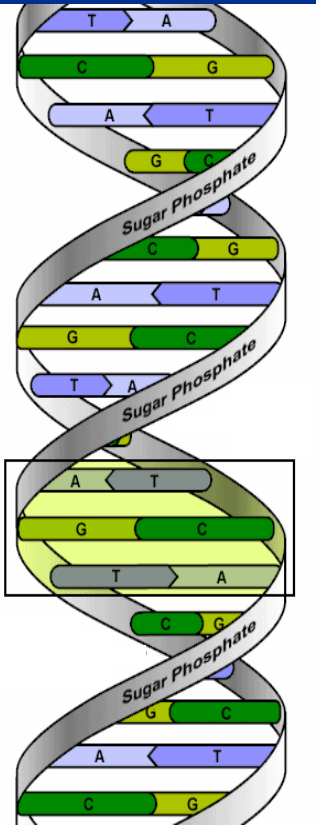
Jungkuk Hur
School of Informatics

October 4, 2005

Overview

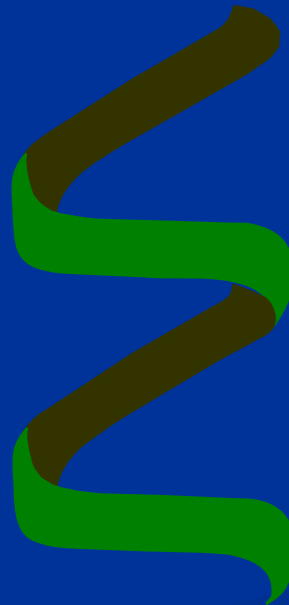
- Introduction to Transcriptional Regulation
- ChIP-on-Chip (ChIP-Chip)
- Current Approaches
- Our Approach

The Central Dogma



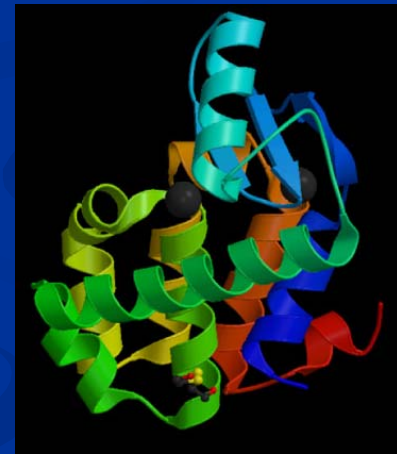
DNA

Transcription



RNA

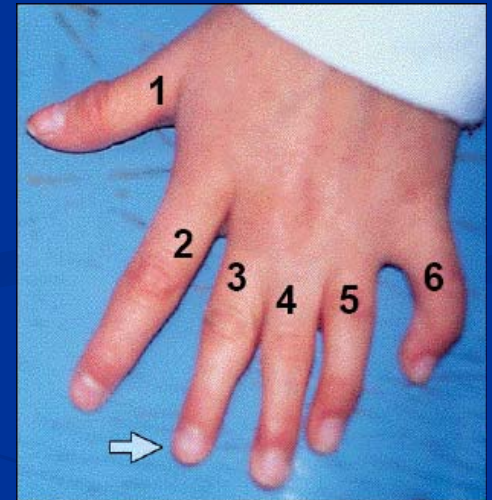
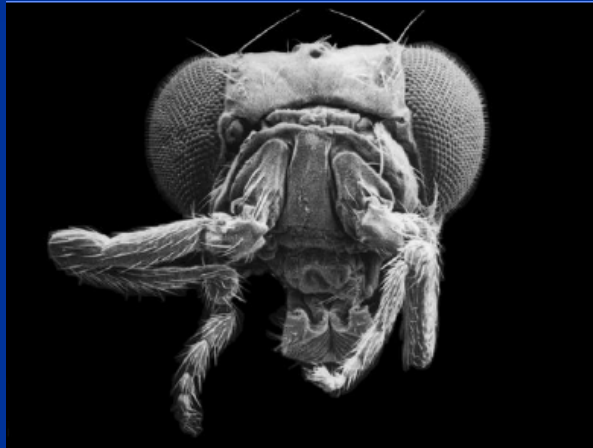
Translation



Protein

Genes need to be regulated

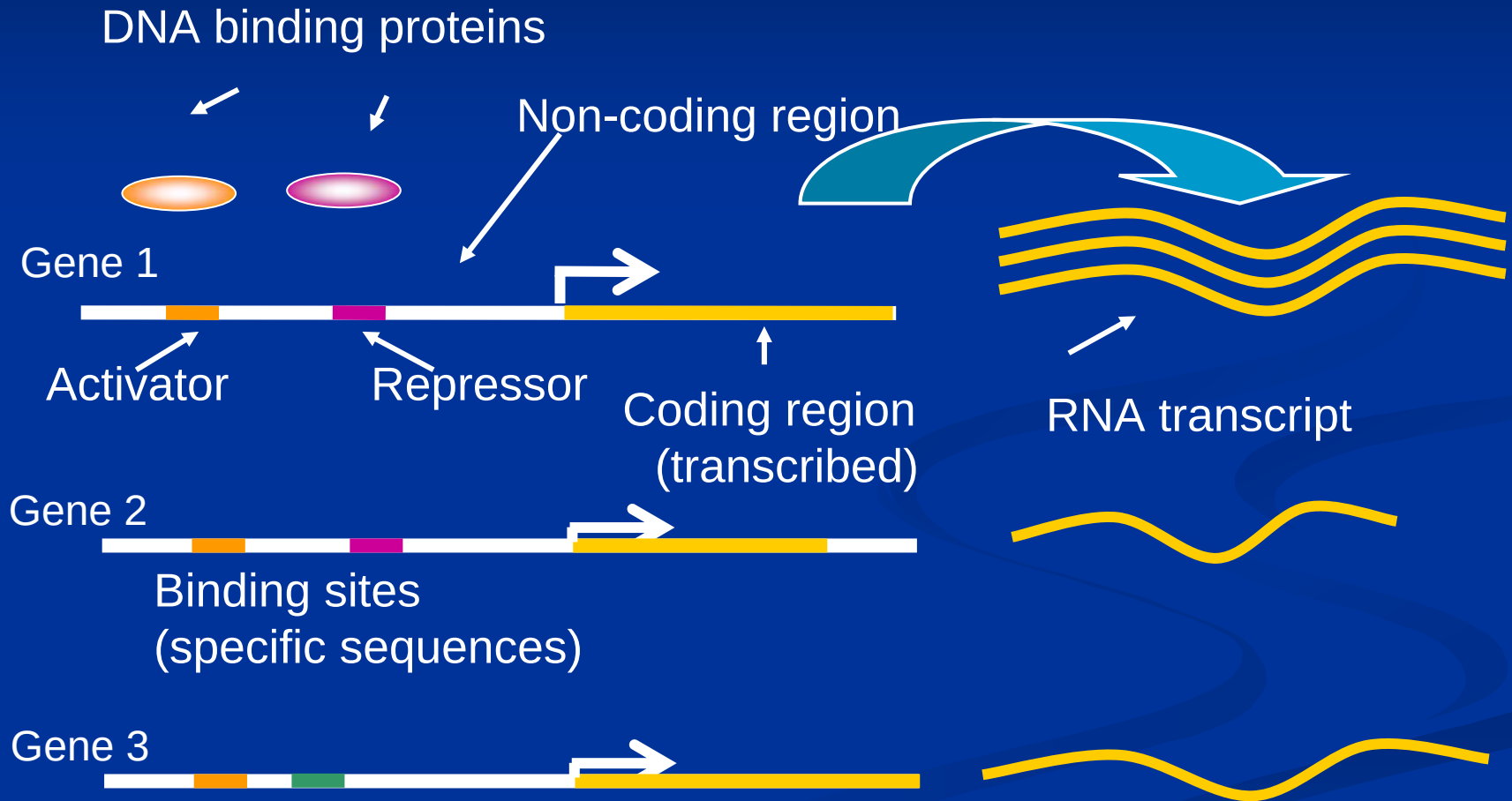
* If gene regulation goes awry?
=> Developmental abnormality



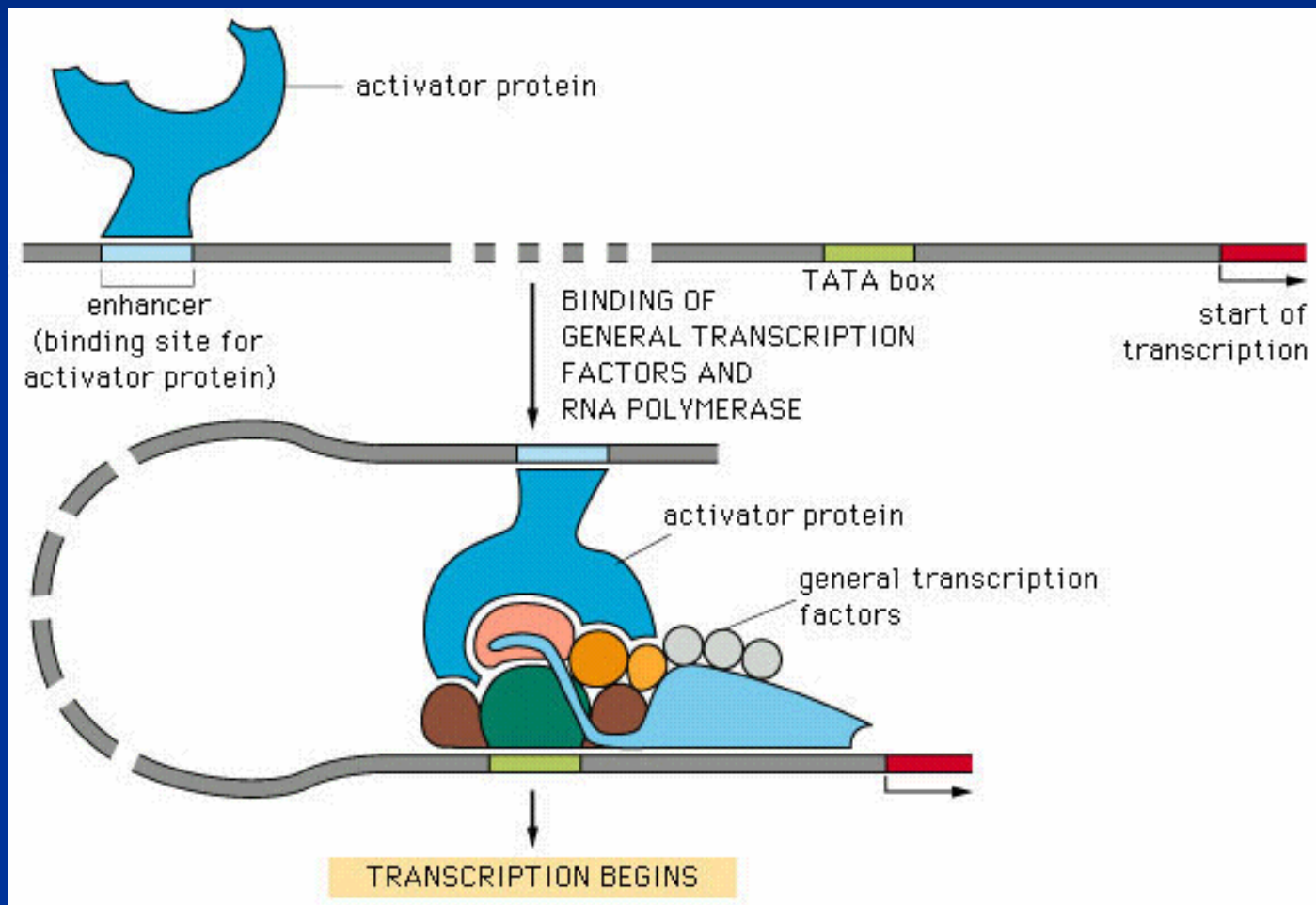
=> Diseases such as
Chronic myeloid leukemia
rheumatoid arthritis

- •transcription
- •post transcription (RNA stability)
- •post transcription (translational control)
- •post translation (not considered gene regulation)
- *usually, when we speak of gene regulation, we are referring to **transcriptional regulation**the “transcriptome”*

Transcriptional Regulation

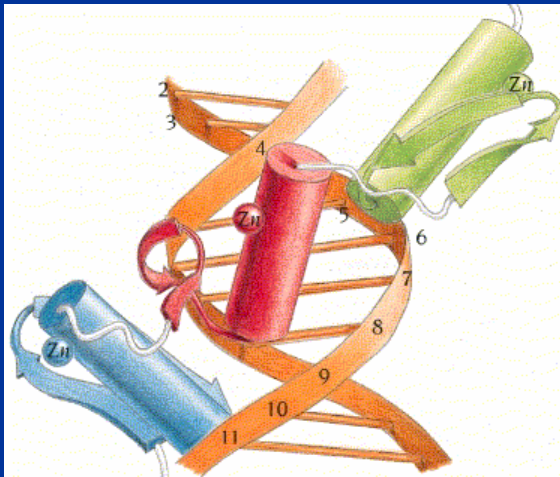


Transcriptional Regulation



Transcription Factor Binding Sites

- Gene regulatory proteins contain structural elements that can “**read**” DNA sequence “**motifs**”
- The amino acid – DNA recognition is not straightforward
- Experiments can pinpoint binding sites on DNA



Zinc finger



Helix-Turn-Helix



Leucine zipper

Modeling Binding Sites

Given a set of (aligned) binding sites ...

- Consensus sequence

- Probabilistic model
(profile of a binding site)

A	4	3	1	1	0	0	1	0	0	1	0	1
C	1	3	0	0	0	0	13	6	0	0	1	9
G	5	5	13	13	14	14	0	8	14	12	13	1
T	4	3	0	0	0	0	0	0	0	1	0	3



GCGGGGCGGGC
 TGGGGGCGGGT
 AGGGGGCGGGG
 TAGGGGCCGGGC
 TGGGGGCGGGT
 TGGGGGCGGGC
 ATGGGGCGGGC
 GTGGGGCGGGC
 AAAGGGCGGGC
 GGGAGGCCGGGA
 GCGGGGCGGGC
 GAGGGGACGAGT
 CCGGGGCGGTCC
 ATGGGGCGGGC

Overview

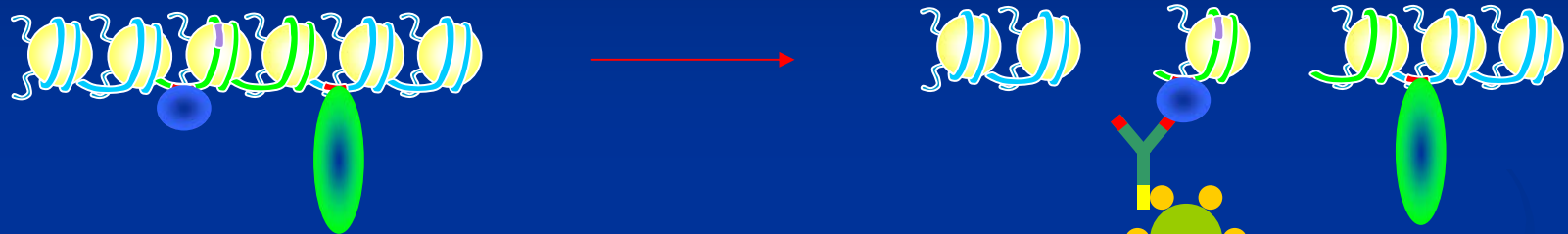
- Introduction to Transcriptional Regulation
- **ChIP-on-Chip (ChIP-Chip)**
- Current Approaches
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ChIP-on-Chip

- Based on
 - ChIP (Chromatin Immuno-Precipitation)
 - Microarray
- In vivo assay
- Genome-wide location analysis

Chromatin Immuno Precipitation (ChIP)

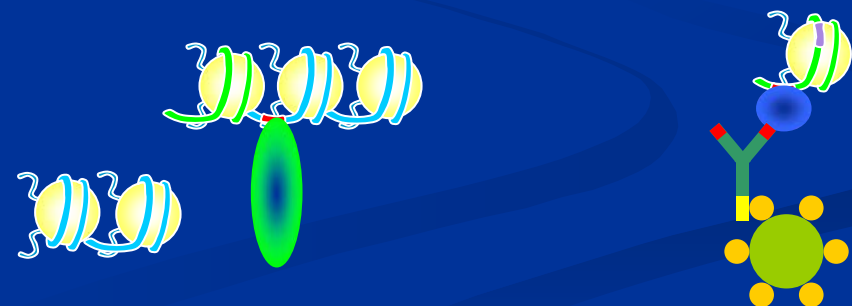
Sonication or
vortexing with glass-beads



Immunoprecipitation

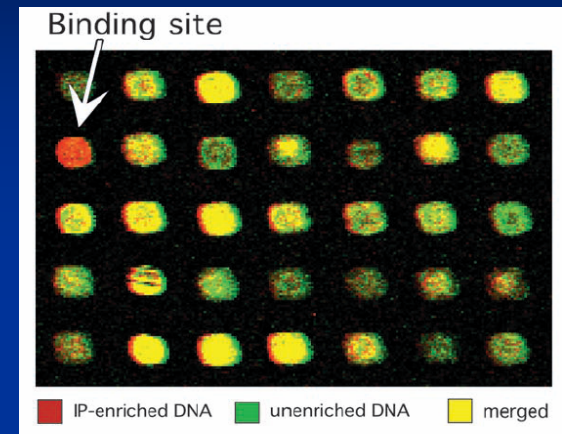
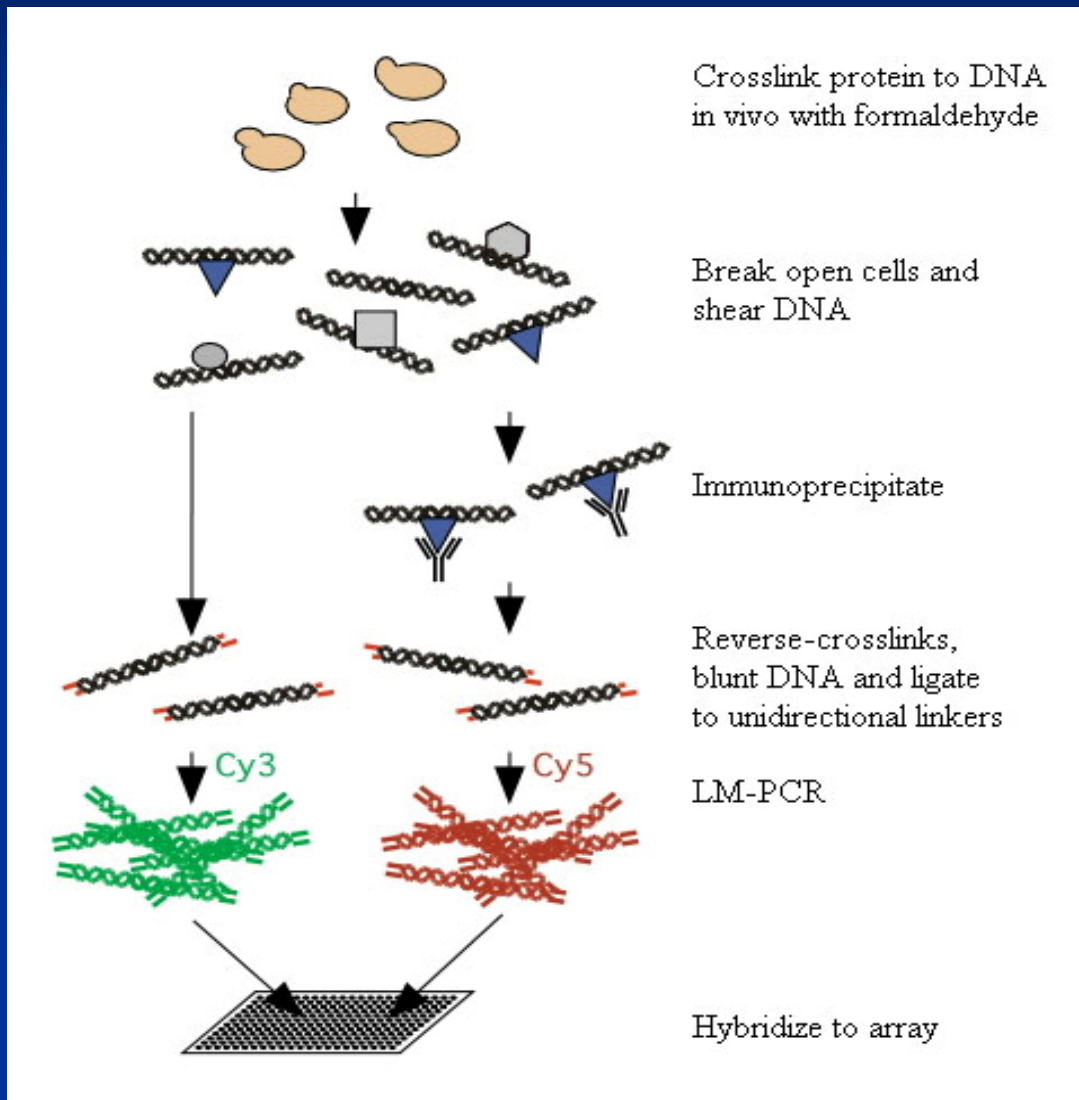
Supernatant

Pellet



- Using antibody of a protein of interest
- DNA bound to specific protein are enriched.

ChIP-on-Chip (Ren *et al.*)



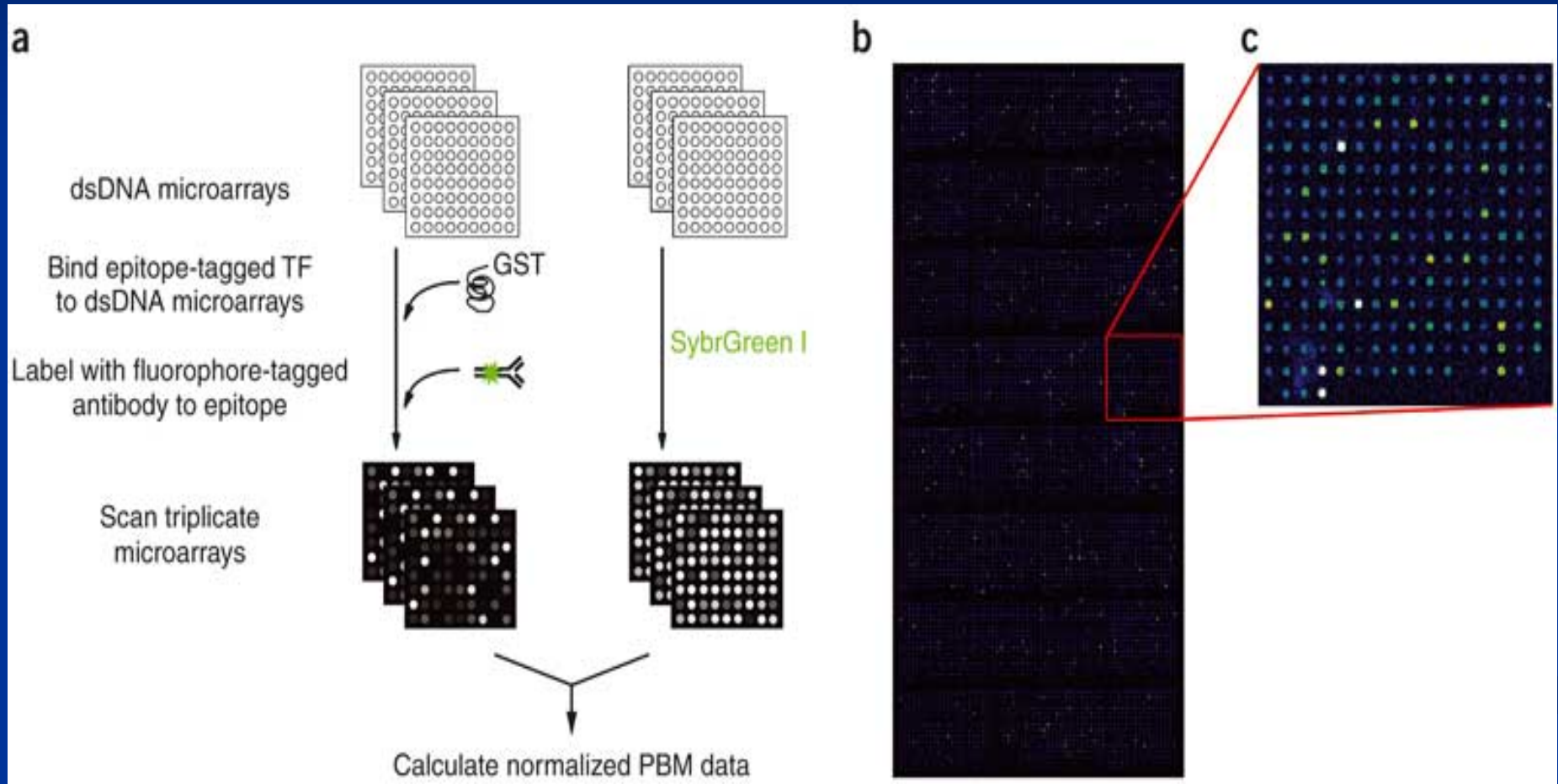
	Galactose	Galactose
Name	p-value	average ratio
iYBR021W	0.000000121	10.3
iYLR080W	0.000000142	9.56
iYBR018C	0.000000038	8.36
iYDR008C	0.000000252	8.26
iYLR094C	0.000000302	7.94
iYBR019C	0.000000696	7.89
iYLR096W	0.00000247	5.4

Array of **intergenic**
sequences from the whole
genome

Protein Binding Microarray (PBM) (Bulyk *et al.*)

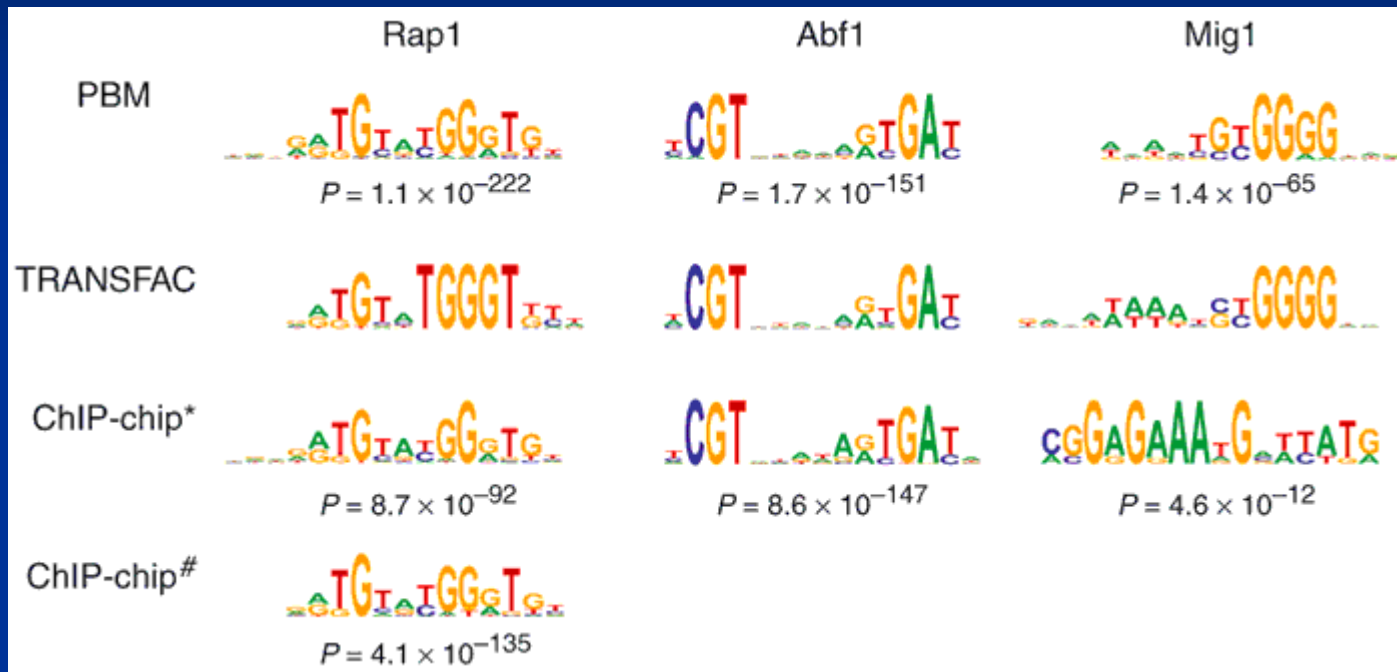
- *In vitro* assay
- DNA-binding protein of interest is expressed with an epitope tag, purified and then bound **directly to a double-strand DNA microarray**
- Can overcome the shortcomings of ChIP-on-Chip
 - Poor enrichment
 - No available antibody
 - Unknown culture condition or time points

Protein Binding Microarray



Whole-genome yeast intergenic microarray bound by Rap1

ChIP-on-Chip vs PBM



- Done by Mukherjee *et al.*
- Useful when ChIP-on-Chip does not result in enough enrichment
- * Lee *et al.* , # Lieb *et al.*

Overview

- Introduction to Transcriptional Regulation
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Approaches

- Representative TFBS (Motif) Discovery
- Understanding Regulatory Modules

Motif Discovery

- MEME (Expectation Maximization)
- CONSENSUS (greedy multiple alignment)
- WINNOWER (Clique finding in graphs)
- SP-STAR (Sum of pairs scoring)
- MITRA (Mismatch trees to prune exhaustive search space)
- BioProspector (Gibbs Sampling Based)
- MDScan (Differential weight for sequences)
- Motif Regressor
- EBMF (Energy Based Motif Finding)

Transcriptional regulatory code by Harbison et al.

- *Saccharomyces cerevisiae* (budding yeast) - Eukaryote
- TFBS binding analysis
- Simple regulatory models
- 203 TFs in rich media +

¹ Highly hyperoxic

² Mildly hyperoxic

³ Amino acid starved

⁴ Nutrient deprived

⁵ Filamentation

⁶ Mating

⁷ Heat

⁸ Galactose

⁹ Raffinose

¹⁰ Acidic

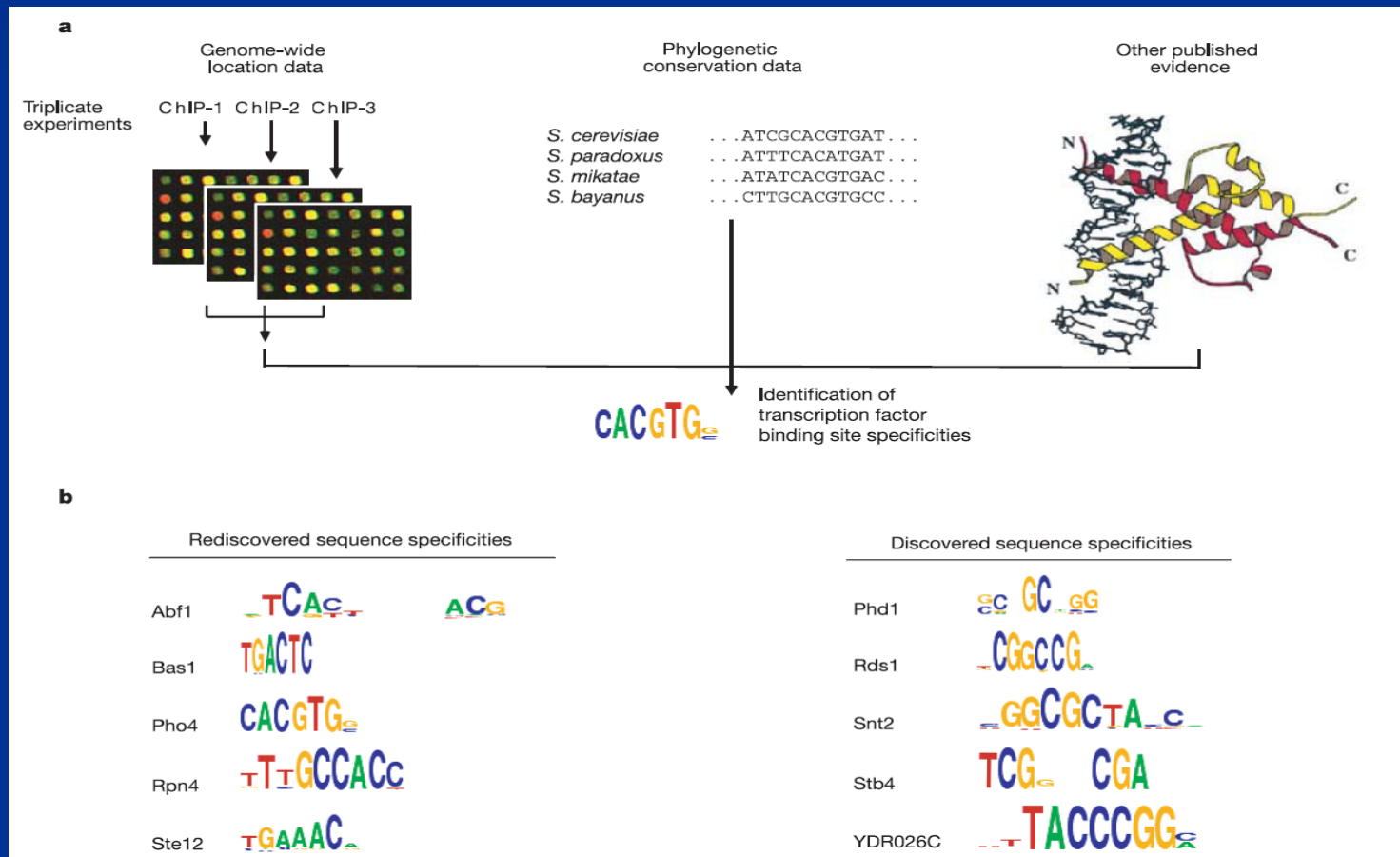
¹¹ Phosphate deprived

¹² Vitamin deprived

- Genome-wide location data 11,000 unique interaction ($p < 0.001$)

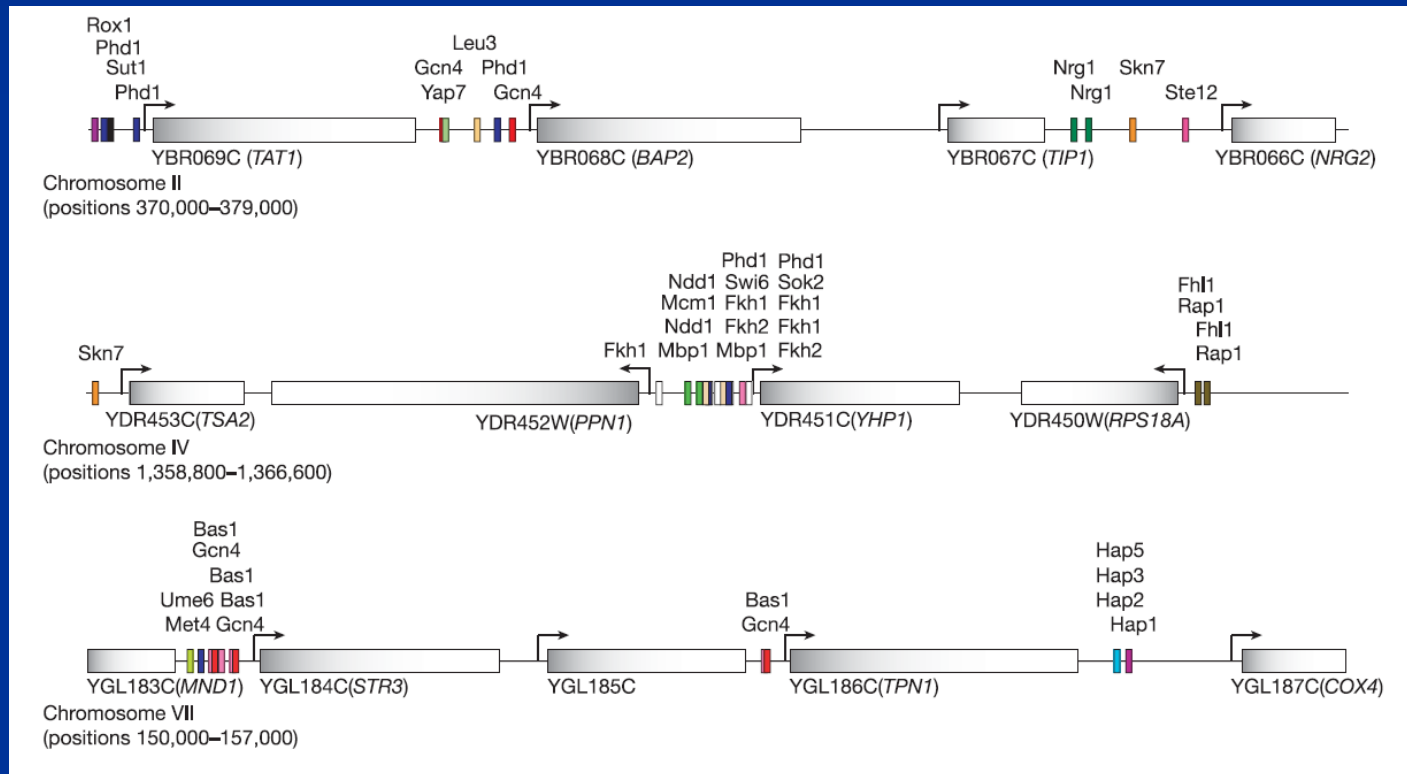
Transcriptional regulatory code by Harbison et al.

■ Identification of transcription factor binding site specificities



Transcriptional regulatory code by Harbison et al.

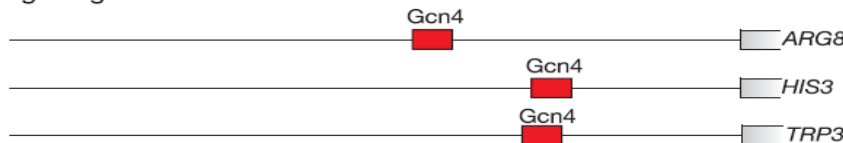
■ Construction of regulatory map of Yeast



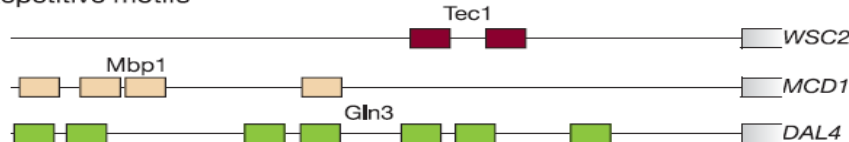
Transcriptional regulatory code by Harbison et al.

■ Promoter architectures

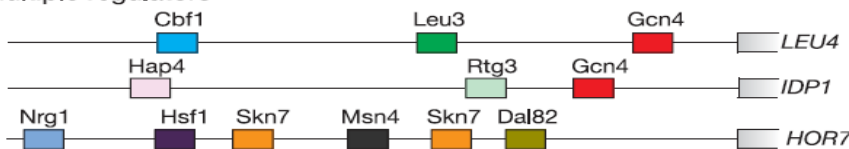
Single regulator



Repetitive motifs



Multiple regulators



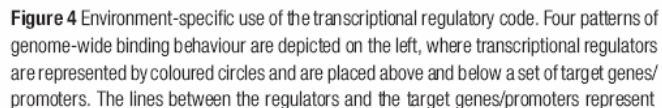
Co-occurring regulators



Supplementary Table 4 Overrepresented MIPS categories among single-regulator architecture binding targets

Regulator	P value ¹	Enriched MIPS category ²	
Bas1	6.10e-09	nucleotide metabolism*	Rim1, Sko1
Fhl1	1.73e-15	ribosome biogenesis	Rox1, Sut1
Gal4	2.18e-04	C-compound and carbohydrate metabolism*	Sip4, Stp1
Gat1	4.92e-05	nitrogen and sulfur metabolism*	Skn7, Sok2
Gat1	2.63e-02	mRNA transcription*	Skn7, Sut1
Gat1	4.38e-02	amino acid metabolism	Skn7, Swi6
Gcn4	8.72e-12	amino acid metabolism*	Skn7, Xbp1
Gzf3	2.21e-02	transport mechanism	Sko1, Sok2
Hap3	6.03e-03	lipid, fatty-acid and isoprenoid metabolism	Sok2, Sut1
Hap3	1.61e-02	allantoin and allantate transporters	Sok2, Swi6
Hap3	2.50e-02	other energy generation activities	Spt23, Yox1
Hap4	3.33e-10	respiration	Stb1, Swi4
Hap4	1.78e-05	mitochondrial transport	Stb1, Swi6
Hap4	1.03e-02	transport mechanism	Stb1, Tec1
Hap4	2.12e-02	assembly of protein complexes	Ste12, Swi4
Hsf1	6.58e-06	stress response*	Ste12, Swi6
Ino4	5.31e-03	lipid, fatty-acid and isoprenoid metabolism*	Ste12, Tec1
Mbp1	1.04e-04	DNA processing	Swi4, Swi6
Met32	1.13e-04	amino acid metabolism*	Swi4, Tec1
Met32	1.21e-03	nitrogen and sulfur metabolism*	Swi6, Tec1
Met32	4.64e-02	amino-acid transporters	Yap1, Yap7
Mot3	3.89e-02	DNA processing	
Msn2	4.40e-02	metabolism of energy reserves (glycogen, trehalose)	
Put3	3.45e-02	other transport facilitators	
Reb1	2.09e-05	vesicular transport (Golgi network, etc.)	
Rfx1	3.57e-02	other protein-synthesis activities	
Rox1	3.43e-02	cell death	
Rpn4	2.49e-13	proteolytic degradation*	
Rtg3	8.50e-03	other transcription activities	
Sig1	2.97e-02	cell cycle	
Sip4	2.69e-03	glyoxylate cycle	
Sip4	1.57e-02	glycolysis and gluconeogenesis	
Stb4	4.02e-02	allantoin and allantate transporters	
Stb5	2.42e-02	electron transport and membrane-associated energy conservation	
Ste12	5.56e-03	cell differentiation*	
Sut1	5.37e-03	glyoxylate cycle	
Swi6	7.96e-03	nitrogen and sulfur metabolism	
Thi2	1.15e-02	mRNA transcription*	
Thi2	2.45e-02	metabolism of vitamins, cofactors, and prosthetic groups	

■ Environment-specific use of regulatory codes



binding events. Specific examples of the environment-dependent behaviours are depicted on the right. Coloured circles represent regulators and coloured boxes represent their DNA binding sequences within specific promoter regions. We note that regulators might exhibit different behaviours when different pairs of conditions are compared.

Overview

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Our Approaches

- Better understanding of differential binding of TF and DNA in different conditions by using ChIP-on-Chip and gene expression data.

Obstacles in TFBS Analysis

- Variation in binding sequences might be problematic in motif discovery process.
 - But for differential binding, there is no sequence discrepancy.
- For eukaryotic systems, lots of transcription factors (TFs) work together with other TFs affecting each other's binding to DNA

Causes of Differential Binding

- We suspect the possible causes for this differential binding to be
 - Changes in the TF expression
 - Changes in other TFs expression
 - Modifications in TFs (protein level)
 - Changes in physical structures (epigenetic features)
 - Other unknown reasons

Cooperations in TFs



Condition 1



Condition 2



Condition 3

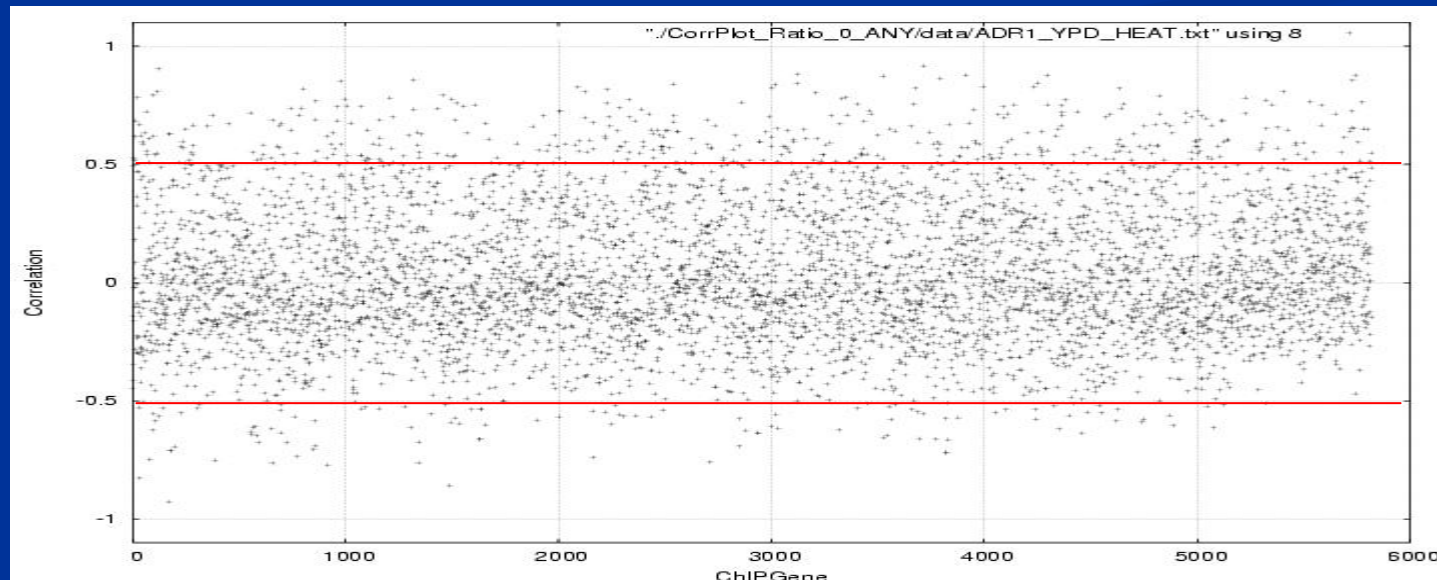
- What has caused the difference in the binding affinity?

Differentially Bound Promoters

■ Simple correlation

$$\frac{A - B}{A + B}$$

(A, B: binding ratio of TF in condition 1 and 2 respectively)



Differentially Bound Promoters

Con1 vs Con2
Gene_1
Gene_2
Gene_3
Gene_4
~
Gene_n

Con1 vs Con3
Gene_2
Gene_5
Gene_7
Gene_8
~
Gene_n

Con2 vs Con3
Gene_1
Gene_2
Gene_3
Gene_4
~
Gene_n

- How can we confirm which other TF(s) is involved?

Methods

- How can we confirm which other TF(s) is involved?
 - Sequence analyses on the differentially bound promoters?
 - Comparison of ChIP-on-Chip results?
 - Protein-protein interaction between TFs?
- Other possible analysis
 - Gene Ontology distribution of differentially bound promoters

Expected Results

- We may be able to use heterogeneous experimental data to reveal the underlying mechanisms of differential binding of transcription factor to cis-regulatory region.

Thank you

Any question and suggestion ?